Waves

Definition of Wave:

If a vibratory disturbance occurs at any point in an elastic medium, this disturbance will be transmitted from one layer to the next through the medium, because of the elastic forces on adjacent layers, but this medium itself doesn't move as a whole. This is called wave.

Types of wave:

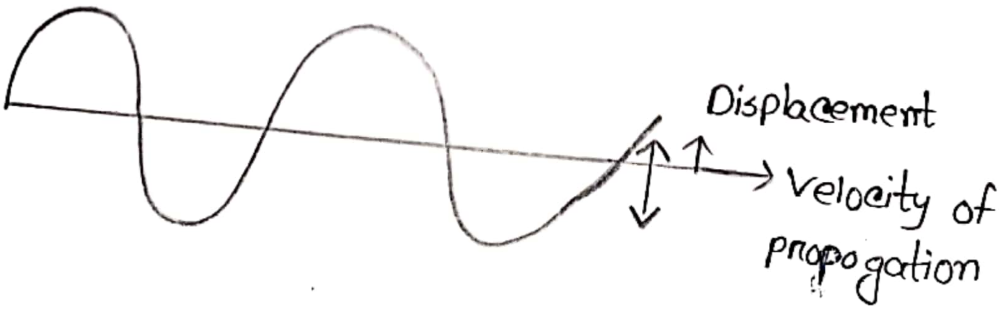
Generally there are two types of waves. Such as:-

① Transverse Wave

② Longitudinal Wave

Transverse Wave:

When the displacement of the medium is perpendicular to the propogation of the wave, then the wave is termed as Transverse Wave.



Longitudinal Wave:

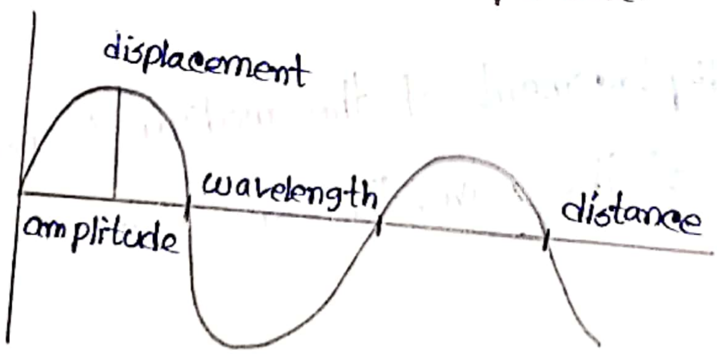
When the displacement of the medium is parallel to the propogation of the wave, then the wave is termed as longitudinal wave.

There are another two types of waves. They are:-

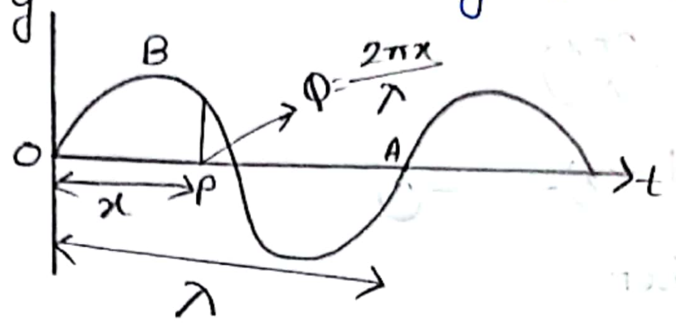
1. Progressive wave
2. Stationary of Standing wave.

Progressive Wave:

Progressive wave is defined as the onward transmission of the vibratory of a body in an elastic medim from one particle to the successive particle.



Differential Equation of Progressive wave:



Plane progressive wave.

Let us assume that a progressive wave travels from the origin O along the positive direction of x ~~axis~~. axis, from left to right. The displacement of a particle at a given instant is,

$$y = a\sin\omega t \text{ ——— (1)}$$

where a is the amplitude of the vibration of the particle and $\omega = 2\pi n$.

The displacement of the particle P at a distance x from O at a given instant is given by,

$$y = a\sin(\omega t - \phi) \text{ ——— (2)}$$

If the two particles are separated by a distance λ, they will differ by a phase of 2π. Therefore, the phase ϕ of the particle P at a distance x is,

$$\phi = \left(\frac{2\pi}{\lambda}\right)x$$

Therefore, the displacement of the wave is,

$$y = a\sin\left(\omega t - \frac{2\pi x}{\lambda}\right)$$

$$= a\sin(\omega t - kx) \quad \text{——— (3)}$$

where, wave number

$$k = \frac{2\pi}{\lambda}$$

Since $\omega = 2\pi n = 2\pi\left(\frac{v}{\lambda}\right)$

The equation is given by,

$$y = a\sin\left[\left(\frac{2\pi vt}{\lambda}\right) - \left(\frac{2\pi x}{\lambda}\right)\right]$$

$$= a\sin\frac{2\pi}{\lambda}(vt - x) \quad \text{——— (4)}$$

If the wave travels in opposite direction, the equation becomes,

$$y = a\sin\frac{2\pi}{\lambda}(vt + x) \quad \text{——— (5)}$$

Now differentiating equation (3) with respect to t. we get,

$$\frac{dy}{dt} = a\omega\cos(\omega t - kx)$$

Differentiating again with respect to t. we get,

$$\frac{d^2y}{dt^2} = -a\omega^2\sin(\omega t - kx)$$

$$= -\omega^2 y \quad \text{—— (6)}$$

Differentiating equation (3) with respect to x. we get,

$$\frac{dy}{dx} = -aK\cos(\omega t - Kx)$$

Differentiating again with respect to x. we get,

$$\frac{d^2y}{dx^2} = -aK^2\sin(\omega t - Kx)$$

$$= -K^2 y \quad \text{—— (7)}$$

From equation (6) and (7) we get,

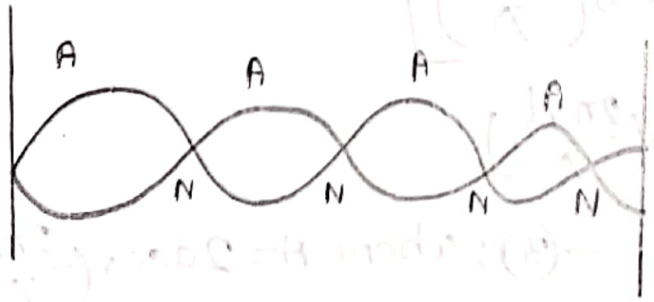
$$\frac{d^2y}{dt^2} = \left(\frac{\omega^2}{K^2}\right) \cdot \left(\frac{d^2y}{dx^2}\right)$$

So, $$\boxed{\frac{d^2y}{dt^2} = v^2\left(\frac{d^2y}{dx^2}\right)}$$

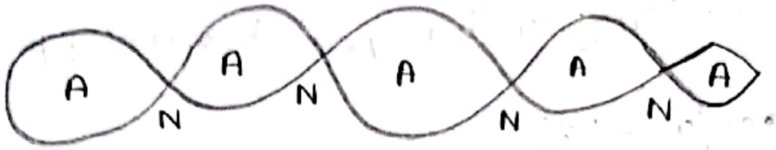
This is the required differential from of progressive wave.

Stationary wave:

Stationary waves are produced when two waves interfere in an invisible disturbance through the wave is moving through out the material.



Equation of stationary wave:



Stationary wave

Let us consider a progressive wave of amplitude a and wavelength λ travelling in the direction of x axis.

$$y_1 = a\sin\frac{2\pi}{\lambda}[vt-x] \quad \text{------} \quad (1)$$

This wave is reflected from a free end and it travels in the negative direction of X axis, then,

$$y_2 = a\sin\frac{2\pi}{\lambda}[vt+x] \quad \text{------} \quad (2)$$

According to principle of superposition, the resultant displacement is,

$$y = y_1 + y_2$$

$$= a\left[\sin\frac{2\pi}{\lambda}(vt-x) + \sin\frac{2\pi}{\lambda}(vt+x)\right]$$

$$= a\left[2\sin\left(\frac{2\pi vt}{\lambda}\right)\cos\left(\frac{2\pi x}{\lambda}\right)\right]$$

$$\Rightarrow y = 2a\cos\left(\frac{2\pi x}{\lambda}\right)\sin\left(\frac{2\pi vt}{\lambda}\right)$$

$$= \boxed{A\sin\left(\frac{2\pi vt}{\lambda}\right)} \quad \text{------} \quad (3); \text{ where } A = 2a\cos\left(\frac{2\pi x}{\lambda}\right)$$

This is the equation of a stationary wave.

Nodes and Antinodes:

(a) At points where $x = 0, \lambda/2, 3\lambda/2$, the values of $\cos\frac{2\pi x}{\lambda} = \pm 1$. So, $A = \pm 2a$. At this point the resultant amplitude is maximum. They are called Antinodes as shown in figure.

(b) At points where $x = \lambda/4, \frac{3\lambda}{4}, \frac{5\lambda}{4}$ ..... the values of $\cos\frac{2\pi x}{\lambda} = 0$.

$\therefore A = 0$. The resultant amplitude is zero at these points. They are called Nodes.

The distance between any two successive antinodes or nodes is equal to $\lambda/2$ and the distance between an antinode and a node is $\frac{\lambda}{4}$.

(c) When $t = 0, \frac{T}{2}, \frac{3T}{2}, 2T$ ...... then $\sin\frac{2\pi t}{T} = 0$, the displacement is zero.

(d) When $t = \frac{T}{4}, \frac{3T}{4}, \frac{5T}{4}$ etc, ... $\sin\frac{2\pi t}{T} = \pm 1$, the displacement is maximum.

Beat:

When two sound waves of different frequency approach the ear, the alternating constructive and destructive interference causes the sound to be alternately soft and loud due to the superposition of sound. This phenomenon is known as beat.

Mathematical Analysis of Beat:

Let us consider two waves of slightly different frequencies n_1 and n_2 ($n_1 \sim n_2 < 10$) having equal amplitude travelling in a medium in the same direction.

At time $t=0$, both waves travel in same phase. The equations of the two waves are,

$$y_1 = a \sin \omega_1 t$$
$$= a \sin (2\pi n_1) t \quad \text{--- (1)}$$

$$y_2 = a \sin \omega_2 t$$
$$= a \sin (2\pi n_2) t \quad \text{--- (2)}$$

When the two waves superimpose, the resultant displacement is given by,

$y = y_1 + y_2$

$\Rightarrow y = a\sin(2\pi n_1)t + a\sin(2\pi n_2)t$ —— ③

Therefore,

$$y = 2a\sin 2\pi\{(n_1+n_2)/2\}t\cos 2\pi\{(n_1-n_2)/2\}t \quad \text{— ④}$$

Substitute $A = 2a\cos 2\pi\{(n_1-n_2)/2\}t$ and $n = (n_1+n_2)/2$ in eqn. ④

$$y = A\sin 2\pi nt.$$

This represents a simple harmonic wave of frequency $n = (n_1+n_2)/2$ and amplitude A which changes with time.

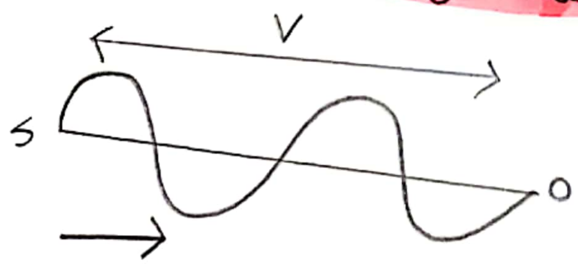
Doppler Effect

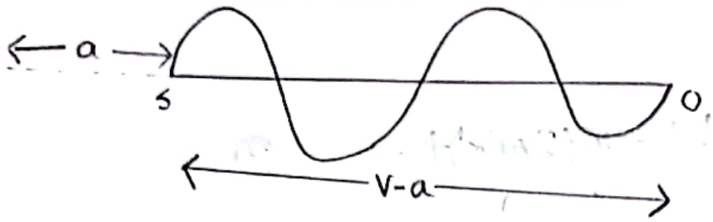
Definition of Doppler Effect:

This apparent change in the pitch due to the relative motion between the source and the observer is called Doppler Effect.

1. Observer at Rest and Source in Motion:

ⓐ When the source moves towards the stationary observer:





Suppose a source S is producing sound of pitch n and wavelength λ. The velocity of sound is V.

Let the source move with a velocity a towards the observer. So, the length is (V-a) and the apparent wavelength,

$$\lambda' = \frac{V-a}{n}$$

we know that,

$$V = n\lambda$$

The apparent pitch,

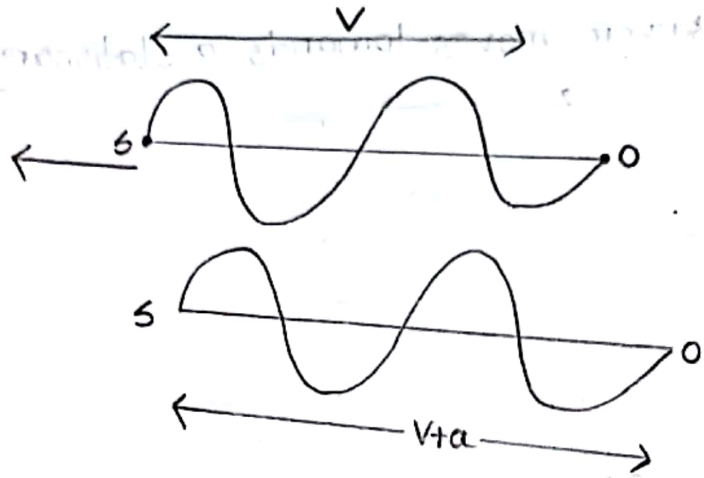
$$n' = \frac{V}{\lambda'}$$

$$\Rightarrow n' = \frac{V}{\frac{V-a}{n}}$$

$$\Rightarrow n' = \left(\frac{V}{V-a}\right)n$$

$\therefore$ The apparent pitch is $\boxed{n' = \left(\frac{V}{V-a}\right)n}$ when the source moves towards a stationary observer.

(b) When the source moves away from the stationary observer:-



Suppose a source S is producing sound of pitch n and wavelength λ. The velocity of sound is V.

Let the source move with a velocity a away from the observer. So, the length is (V+a) and the apparent wavelength,

$$\lambda' = \frac{V+a}{n}$$

The apparent pitch,

$$n' = \frac{V}{\lambda'}$$

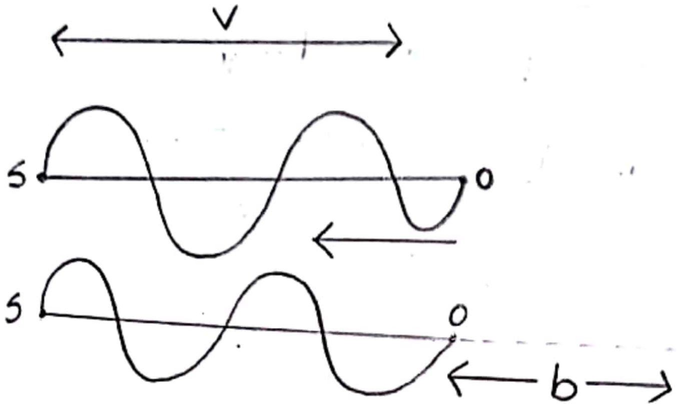
$$= \frac{V}{\frac{V+a}{n}}$$

$$\therefore n' = \left(\frac{V}{V+a}\right)n$$

So, The apparent pitch is $\boxed{n' = \left(\frac{V}{V+a}\right)n}$ when the source moves away from a stationary observer.

2. Source at Rest and Observer in Motion:

ⓐ When the observer moves towards a stationary source:



Suppose a source S is producing sound of pitch n and wavelength λ. The velocity of sound is V.

Let the observer move with a velocity b towards a stationary source. In this case, the apparent wavelength remains the same. The apparent frequency,

$$n' = n + \frac{b}{\lambda}$$

$$\Rightarrow n' = \frac{V}{\lambda} + \frac{b}{\lambda}$$

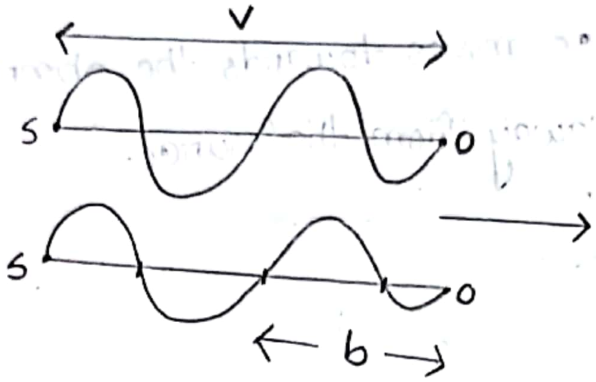
$$\Rightarrow n' = \frac{V+b}{\lambda}$$

But, $\lambda = \frac{V}{n}$

$$\therefore n' = \left(\frac{V+b}{V}\right)n$$

So, The apparent frequency is $\boxed{n' = \left(\frac{V+b}{V}\right)n}$ when the observer moves towards a stationary source.

(b) When the observer moves away from a stationary source:-



Suppose a source S is producing sound of pitch n and wavelength λ. The velocity of sound is V.

Let the observer moves ~~away~~ a velocity b away from a stationary source. In this case, the apparent wavelength remains the same. The apparent frequency,

$$n' = n - \frac{b}{\lambda}$$

$$\Rightarrow n' = \frac{V}{\lambda} - \frac{b}{\lambda}$$

$$\Rightarrow n' = \frac{V-b}{\lambda}$$

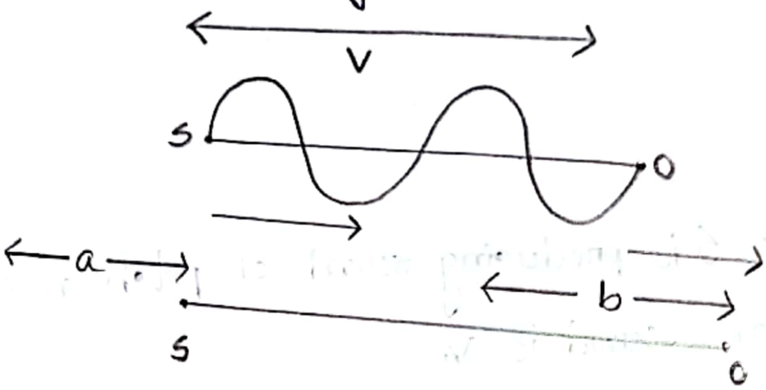
But, $\lambda = \frac{V}{n}$

$$\therefore n' = \left(\frac{V-b}{V}\right)n$$

So, The apparent frequency is $\boxed{n' = \left(\frac{V-b}{V}\right)n}$ when the observer moves away from a stationary source.

③ When both the Source and the observer are in Motion.

When the source moves towards the observer and the observer moves away from the source.



Suppose a source S is producing sound of pitch n and wavelength λ. The velocity of sound is V. The velocity of the source is a and the velocity of the observer is b.

Let the source moves towards the observer with a velocity a and the observer moves away from the source with a velocity b.

The apparent wavelength,

$$\lambda' = \frac{(v-a)}{n}$$

and, $n' = \left(\frac{v-b}{\lambda'}\right)$

$$\boxed{\therefore n' = \left(\frac{v-b}{v-a}\right) n} \quad \text{--- (1)}$$

Special Cases:

(a) When the source and observer move towards each other.

In equation (1) we get,

$$n' = \left(\frac{v-b}{v-a}\right)n$$

Taking b negative in this equation.

$$n' = \left[\frac{v-(-b)}{v-a}\right]n$$

$$\therefore \boxed{n' = \left(\frac{v+b}{v-a}\right)n}$$

(b) When the source and observer moves away from each other.

In equation (1) taking a to be negative,

$$\therefore \boxed{n' = \left(\frac{v-b}{v+a}\right)n}$$

(c) Source moving away from the observer and the observer moving towards the source.

In equation (1) taking both a and b negative.

$$n' = \left[\frac{v-(-b)}{v-(-a)}\right]n$$

$$\therefore \boxed{n' = \left[\frac{v+b}{v+a}\right]n}$$

Beat

Condition for loud (maxima) and soft (minima):

1. The resultant amplitude is maxima (i.e) $= \pm 2a$, if

$\cos 2\pi[(n_1-n_2)/2] \cdot t = \pm 1$

so $2\pi[(n_1-n_2)/2]t = 0, \pi, 2\pi, \ldots\ldots, m\pi$

or, $[(n_1-n_2)/2]t = 0, 1, 2, \ldots\ldots, m.$

The first maximum is obtained at $t_1 = 0$

The second maximum is obtained at,

$t_2 = 1/n_1-n_2$

The third maximum at $t_3 = \frac{2}{n_1-n_2}$ and so on.

The time interval between two successive maxima is.

$t_2-t_1 = t_3-t_2 = \frac{1}{n_1-n_2}$

Hence, the number of beats produced per second is equal to the reciprocal of the time interval between two successive maxima.

2. The resultant amplitude is minimum (i.e) equal to zero, if

$\cos 2\pi[(n_1-n_2)/2]t = 0$

(i.e) $2\pi[(n_1-n_2)/2]t = \pi/2, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots, (2m+1)\pi/2$

$\Rightarrow [(n_1-n_2)/2]t = \frac{1}{2}, \frac{3}{2}, \frac{5}{2} \ldots (2m+1)/2$

Where $m = 0, 1, 2 \ldots$

The first minimum is obtained at,

$$t_1' = \frac{1}{2(n_1-n_2)}$$

The second minimum is obtained at.

$$t_2' = \frac{3}{2(n_1-n_2)}$$

The third minimum is obtained at.

$$t_3' = \frac{5}{2(n_1-n_2)}$$ and so on.

Time interval between two successive minima is.

$$t_2' - t_1' = t_3' - t_2' = \frac{1}{n_1-n_2}$$

Hence, the number of beats produced per second is equal to the reciprocal of time interval between two successive minima.

Group Velocity:

The group velocity of a wave is the velocity with which the overall shape of the wave's amplitude propagates through space.

The group velocity V_g is defined by the equation:

$$V_g = d\omega/dK$$

Where, ω is the wave's angular frequency and K is the angular wavenumber.

Phase Velocity:

The phase velocity of a wave is the rate at which the phase of the wave propagates in space.

$$V_p = \omega/K.$$

Relation between Group Velocity and Phase Velocity:

We know that,

Phase velocity, $v_p = \omega/K$ —— (1)

Again, energy of the wave is,

$$E = h\nu = h/2\pi \cdot (2\pi\nu)$$

$$= \hbar\omega$$

and momentum, $p = \frac{h}{\lambda} = (h/2\pi)(2\pi/\lambda) = \hbar K$

So, equation (1) becomes,

$$v_p = \hbar\omega/\hbar p = E/p \text{ —— (2)}$$

For relativistic particle,

$$E = mc^2$$

$$P = mv$$

Now equation (2) becomes,

$$v_p = mc^2/mv = c^2/v \text{ —— (3)}$$

Group velocity,

$$v_g = d\omega/dK$$

$$= d(\hbar\omega)/d(\hbar P)$$

$$= dE/dP \text{ —— (4)}$$

Again for relativistic particle,

$$E^2 = m_0^2 c^4 + p^2 c^2 \quad \text{——— (5)}$$

Differenciation above equation,

$$2E\,dE = 0 + 2pc^2\,dp$$

$$\Rightarrow dE/dp = pc^2/E = v$$

Hence equation (4) becomes,

$$v_g = v$$

Therefore, from equation (4) we get,

$$v_p = c^2/v_g$$

$$\boxed{\Rightarrow v_g v_p = c^2}$$

This is the relation between group velocity and phase velocity.

Math

Example-4.1

If the frequency of a tuning fork is 400 and the velocity of sound in air is 320 m/s, find how far sound travels while the fork completes 30 Vibrations.

Solution:

Here, $n=400$, $V=320\ m/s$, $\lambda=?$

we know that,

$V=n\lambda$

$\Rightarrow \lambda=\frac{V}{n}=\frac{320}{400}=0.8$ meter

$\therefore$ Distance travelled by the wave when the fork completes 1 vibration $=0.8m$

Distance travelled by the wave when the fork completes 30 vibrations $=0.8\times30=24$ meters

Ans:

Example-6.3

A tunning fork A produces 4 beats/second with a tuning fork B of frequency 256. A is ~~failed~~ filed and the beats occur at shorter intervals. What was its original frequency?

Solution:

Frequency of $B = 256$

Beats/s $= 4$

Frequency of $A = 256 + 4 = 260$

or, $256 - 4 = 252$

When A is filed its frequency increases. Suppose the original frequency of A is 260. When it is filed its frequency will be more than 260 and beats/second increase i.e. the beats occur at shorter intervals, which is given.

Hence the original frequency of $A = 260$

Ans:

Example- 6.4

A note produces 4 beats/second with a tuning fork of frequency 512 and 6 beats/second with a tuning fork of frequency 514. Find the frequency of the note.

Solution:

In the first case,

Frequency of a tuning fork $= 512$

Beats/second $= 4$

∴ possible frequencies of the note are $512+4=516$
$512-4=508$

In the second case,

Frequency of the tunning fork $=514$

Beats/second $=6$

possible frequencies of the note are $514+6=520$
$514-6=508$

or, Hence the frequency of the note is 508.

Ans:

Example - 6.5

Two tuning forks A and B give 5 beats/second. The frequency of $A=512$. When B is filed, 5 beats/second are again produced. Find the frequency of B before and after filing.

Solution:

Frequency of $A=512$

Beats per second $=5$

Frequency of B before filling is either,

$512+5=517$

$512-5=507$

or, Since after filing 5 beats/second are again produced, frequency of B after filing is either

$512+5=517$

or, $512-5=507$

Let us consider the frequency of B before filling as 517. After filling the frequency increases. Therefore, after filling the frequency of B can not be equal to either 517 or 507. Therefore frequency of B cannot be equal to 517.

Consider the frequency of B before filing $= 507$

After filing its frequency can be equal to 517.

Therefore, this is the possible value.

Hence, frequency of B before filing $= 507$

Hence, frequency of B after filing $= 517$.

Ans:

Example-6.7

A turning fork A of frequency 384 Hz gives 6 beats per second when sounded with another turning fork B. On loading B with a little wax, the number of beats per second becomes 4. What is the frequency of B?

Solution:

Frequency of A = 384 Hz

Beats per second = 6

Frequency of B before loading is either 384+6 = 390 Hz

or, 384−6 = 378 Hz

After loading, beats per second = 4

Frequency of B after loading is either, 384+4 = 388 Hz

384−4 = 380 Hz

Since, After loading frequency decreases, the frequency of B before loading is 390 Hz.

Ans:

CamScanner

Example-91

A person is standing on a railway platform. An engine while approaching the platform blows a whistle of pitch 660 hertz. The speed of the engine is 72 Km/hr, velocity of sound = 350 m/s. Calculate the apparent pitch of the whistle as heard by the person.

Solution:

Given that,

$V = 350\ m/s$

$a = 72\ Km/hr = 20\ m/s$

$n = 660$ hertz

The source is moving towards a stationary observer.

We know that,

$$\boxed{n' = \left(\frac{V}{V-a}\right)n}$$

$$= \left(\frac{350}{350-20}\right) \times 660$$

$\therefore n' = 700$ hertz

Ans;

Example-9.2

A person is standing on a platform. A railway engine moving away from the person with a speed of 72 Km/hr blows a whistle of pitch 740 hertz. Calculate the apparent pitch of the whistle as heard by the person. The velocity of sound =350m/s.

Solution:

Given that,

$V = 350 m/s$

$a = 72 Km/hr = 20 m/s$

$n = 740$ hertz

The source is moving away from a stationary observer.

so, we know that,

$$\boxed{n' = \left(\frac{V}{V+a}\right)n}$$

$$\Rightarrow n' = \left(\frac{350}{350+20}\right) \times 740$$

$$\therefore n' = 700 \text{ hertz.}$$

Ans:

Example-9.3

A person is standing near a railway track and a train moving with a speed of 36 km/hr is approaching him. The apparent pitch of the whistle as heard by the person is 700 hertz. Calculate the actual frequency of the whistle. Velocity of sound = 350m/s.

Solution:

Given that,

$V = 350 m/s$

$a = 36 \ km/hr = 10 m/s$

$n' = 700$ hertz

$n = ?$

The source is moving towards a stationary observer.

$$\therefore \boxed{n' = \left(\frac{V}{V-a}\right) n}$$

$$\Rightarrow n = n' \left(\frac{V-a}{V}\right)$$

$$\Rightarrow n = 700 \left(\frac{350-10}{350}\right)$$

$$= 680 \text{ hertz.}$$

Ans:

Example-9.4

Two aeroplanes A and B are approaching each other with a speed of 360Km/hr. The frequency of the whistle emitted by A is 1000 hertz. Calculate the apparent pitch of the whistle as heard by the passengers of aeroplane B. velocity of sound in air = 350 m/s.

Solution:

Given that,

$V = 350\,m/s$

$a = 360\,Km/hr = 100\,m/s$

$b = 360\,Km/hr = 100\,m/s$

$n = 1000$ hertz.

The source and the observer are moving towards each other.

we know that,

$$\boxed{n' = n\left[\frac{V+b}{V-a}\right]}$$

$$\Rightarrow n' = 1000\left(\frac{350+100}{350-100}\right)$$

$$\Rightarrow n' = 1800 \text{ hertz}.$$

Ans:

Example-9.5

Two aeroplanes A and B are moving away from one another with a speed of 720 km/hr. The frequency of the whistle emitted by A is 1100 hertz. Calculate the apparent ~~fr~~ frequency of the whistle as heard by the passengers of the aeroplane B. velocity of sound in air = 350 m/s.

Solution:

Given that,

$V = 350 m/s$

$a = 720 Km/hr = 200 m/s$

$b = 720 Km/hr = 200 m/s$

$n = 1100$ hertz

The source and the observer are moving away from one another.

$$\therefore \boxed{n' = \left[\frac{V-b}{V+a}\right] n}$$

$$\Rightarrow n' = \left[\frac{350-200}{350+200}\right] \times 1100$$

$$\therefore n' = 300 \text{ hertz.}$$

Ans:

Example-9.6

Two aeroplanes A and B are approaching each other and their velocities are 108 Km/hr ~~res~~ and 144 Km/hr respectively. The frequency of a note emitted by A as heard by the passengers in B is 1170 hertz. Calculate the frequency of the note heard by the passengers in A. Velocity of sound = 350m/s.

Solution:

Given that,

$V = 350 m/s$

$a = 108 \ Km/hr = 30 m/s$

$b = 144 \ Km/hr = 40 m/s$

$n' = 1170$ hertz

$n = ?$

The source and the observer are approaching each other.

$$\therefore \boxed{n' = \left(\frac{V+b}{V-a}\right) n}$$

$$\Rightarrow 1170 = \left(\frac{350+40}{350-30}\right) \times n$$

$$\Rightarrow n = \frac{1170 \times 320}{390}$$

$$= \cancel{960 \text{ hertz.}} \ 1434.19 hz$$

Ans:

Example-27

Two aeroplanes pass each other in opposite directions and one of them is blowing a whistle of frequency 540 hertz. Calculate the frequencies of the notes heard in the other aeroplane (1) before and (11) after they have passed each other. Velocity of either of the aeroplanes is 540 Km/hr and the velocity of sound = 350 ms^{-1}.

Solution:

Given that,

$V = 350\ m/s$

$a = 540\ Km/hr = 150\ m/s$

$b = 540\ Km/hr = 150\ m/s$

$n = 540$ hertz.

(1) In the first case, the source and the observer are moving towards each other. Therefore,

$$\boxed{n' = n\left[\frac{V+b}{V-a}\right]}$$

$$\Rightarrow n' = 540\left(\frac{350+150}{350-150}\right)$$

$$\therefore n' = 1350 \text{ hertz}$$

(ii) In the second case, the source and the observer are moving away from each other.

$$\therefore \boxed{n' = \left(\frac{v-b}{v+a}\right)n}$$

$$\Rightarrow n' = \left(\frac{350-150}{350+150}\right) \times 540$$

$$\Rightarrow n' = 216 \text{ hertz}$$

Ans:

Example-9.9

A car emitting sound of frequency 200 hertz is moving away from a stationary observer and towards a rigid flat wall. The velocity of the car is 5 m/s. Calculate the number of beats heard per second by the observer. Velocity of sound in air = 350 m/s.

Solution:

Here, $n = 200$ hertz

$V = 350$ m/s

Velocity of the car, $a = 5$ m/s

The observer hears sound of apparent frequency,

$$n_1 = \left(\frac{v}{v+a}\right)n$$

$$\Rightarrow n_1 = \left(\frac{350}{350+5}\right) \times 200 = 197.2 \text{ hertz}$$

The observer also hears sound of apparent pitch n_2 due to the waves reflected by the wall. It means the source is moving towards the stationary observer.

$$\therefore n_2 = \left(\frac{v}{v-a}\right) n$$

$$\Rightarrow n_2 = \left(\frac{350}{350-5}\right) \times 200$$

$$\Rightarrow n_2 = 203 \text{ Hz}$$

Therefore, the number of beats produced in one second is given by,

$$n_2 - n_1 = 203 - 197.2$$

$$= 5.8$$